THE INVARIANT DIFFERENTIAL

In this lecture, we study the distinguished differential on an elliptic curve and its compatibility with the group law. We have already encountered this differential while discussing Weierstrass equations and divisors of differentials. The point of the present lecture is to explain why it is called invariant and to see how it gives useful information about isogenies, multiplication maps, and Frobenius.

Throughout, K is a perfect field with fixed algebraic closure $\bar{K}$. Unless otherwise stated, elliptic curves are defined over K, while points and isogenies may be taken over $\bar{K}$.

1. The invariant differential

Let E be given by a Weierstrass equation

$$E: \quad y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6. \tag{1.1}$$

Put

$$F(x,y) = y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6.$$

Since $F(x,y) = 0$ in the function field $K(E)$, differentiating the equation gives

$$F_x\,dx + F_y\,dy = 0,$$

or, more explicitly,

$$(2y + a_1x + a_3)\,dy = (3x^2 + 2a_2x + a_4 - a_1y)\,dx.$$

Thus the two expressions

$$\omega_E = \frac{dx}{2y + a_1x + a_3} = \frac{dy}{3x^2 + 2a_2x + a_4 - a_1y} \tag{1.2}$$

agree wherever both are defined.

Remark 1.1 (Why this formula is natural). A smooth plane curve is given locally by one equation in two variables. Thus the two apparent differentials dx and dy must satisfy one relation, namely

$$F_x\,dx + F_y\,dy = 0.$$

At a point where $F_y \neq 0$, the variable x is a local coordinate, while at a point where $F_x \neq 0$, the variable y is a local coordinate. The relation above gives

$$\frac{dx}{F_y} = -\frac{dy}{F_x}.$$

Therefore the partial derivatives tell us exactly how to pass from the x-coordinate description of the differential to the y-coordinate description. There is also a useful geometric way to think about this. The coordinate x fails to be a local parameter precisely at the ramification points of the map

$$x : E \longrightarrow \mathbb{P}^1.$$

At such a point dx vanishes, but F_y vanishes to the corresponding order, and the quotient dx/F_y remains regular. The second expression $-dy/F_x$ makes this cancellation visible whenever the first expression appears problematic.

Proposition 1.2. *The expressions in* (1.2) *patch together to give a regular nowhere-vanishing differential on all of* E.

Proof. We first work on the affine part $E \setminus \{O\}$. Let

$$U_x = \{P \in E : F_y(P) \neq 0\}, \qquad U_y = \{P \in E : F_x(P) \neq 0\}.$$

On U_x, define

$$\omega_x = \frac{dx}{F_y} = \frac{dx}{2y + a_1x + a_3},$$

and on U_y, define

$$\omega_y = -\frac{dy}{F_x} = \frac{dy}{3x^2 + 2a_2x + a_4 - a_1y}.$$

The denominators are units in the appropriate local rings, so these are regular differentials on U_x and U_y. On the intersection $U_x \cap U_y$, the equality $dF = 0$ gives

$$\omega_x = \omega_y.$$

Since E is smooth, F_x and F_y do not vanish simultaneously at any affine point. Hence

$$U_x \cup U_y = E \setminus \{O\},$$

and the two local expressions patch to a regular differential on the affine part of E.

They are also nonzero there. Indeed, if $F_y(P) \neq 0$, then $x - x(P)$ is a uniformizer at P, and ω_E is a unit multiple of $d(x - x(P))$. Similarly, if $F_x(P) \neq 0$, then $y - y(P)$ is a uniformizer and the second expression shows that ω_E is nonzero at P.

It remains to check the point at infinity $O = [0 : 1 : 0]$.

The projective Weierstrass equation is

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3.$$

On the affine chart $Y = 1$, put

$$u = \frac{X}{Y}, \qquad v = \frac{Z}{Y}.$$

Then O corresponds to $(u, v) = (0, 0)$, and the equation becomes

$$G(u, v) = v + a_1uv + a_3v^2 - u^3 - a_2u^2v - a_4uv^2 - a_6v^3 = 0.$$

It follows that $\mathfrak{m}_O = (u, v) = (u)$ by considering orders of poles. Hence u is a uniformizer at O.

On this chart we have

$$x = \frac{u}{v}, \qquad y = \frac{1}{v}.$$

Therefore

$$dx = \frac{v\,du - u\,dv}{v^2}$$

and

$$2y + a_1x + a_3 = \frac{2 + a_1u + a_3v}{v}.$$

Thus

$$\omega_E = \frac{v\,du - u\,dv}{v(2 + a_1u + a_3v)}.$$

Differentiating $G(u, v) = 0$ gives

$$G_u\,du + G_v\,dv = 0, \qquad \text{so} \qquad dv = -\frac{G_u}{G_v}\,du.$$

Using the equation $G = 0$, a direct computation gives

$$vG_v + uG_u = -v(2 + a_1u + a_3v).$$

Consequently,

$$\omega_E = -\frac{du}{G_v} = -\frac{du}{1 + a_1u + 2a_3v - a_2u^2 - 2a_4uv - 3a_6v^2}.$$

The denominator is a unit at O, so ω_E is regular and nonzero at O.

Thus ω_E is regular and nowhere vanishing on all of E. In particular,

$$\operatorname{div}(\omega_E) = 0.$$

□

Remark 1.3. Under an admissible change of Weierstrass coordinates

$$x = u^2x' + r, \qquad y = u^3y' + su^2x' + t, \qquad u \in K^*,$$

the corresponding differentials satisfy $\omega_E = u^{-1}\omega_{E'}$.

For $Q \in E(\bar{K})$, let

$$\tau_Q : E \longrightarrow E, \qquad P \longmapsto P + Q$$

be translation by Q. Recall that τ_Q is an automorphism of the underlying curve with inverse τ_{-Q}. Unless $Q = O$, it is not a group automorphism, since it does not carry O to O.

Proposition 1.4 (Translation invariance)**.** *Let E be an elliptic curve and let ω be a nonzero regular differential on E. Then*

$$\tau_Q^* \omega = \omega$$

for every $Q \in E(\bar{K})$.

Sketch of proof. It is enough to prove the assertion for the differential in (1.2), since every regular differential is a scalar multiple of it. Fix a point Q, and write

$$P = (x, y), \qquad P + Q = (x_Q, y_Q)$$

on a dense affine open subset on which the addition formulas are defined. A direct differentiation of those formulas, followed by the use of the Weierstrass equation, gives

$$\frac{dx_Q}{2y_Q + a_1 x_Q + a_3} = \frac{dx}{2y + a_1 x + a_3}.$$

This is precisely the equality $\tau_Q^* \omega_E = \omega_E$ on a dense open subset of E. Since both sides are regular differentials on all of E, they agree everywhere. □

2. Addition of maps and pullback of differentials

Let

$$\mu : E \times E \longrightarrow E, \qquad (P, Q) \longmapsto P + Q$$

be the addition morphism, and let

$$\mathrm{pr}_1, \mathrm{pr}_2 : E \times E \longrightarrow E$$

be the two projection maps. Translation invariance leads to the fundamental identity

$$\mu^* \omega = \mathrm{pr}_1^* \, \omega + \mathrm{pr}_2^* \, \omega. \tag{2.1}$$

One should think of this as the elliptic-curve analogue of

$$d(x + y) = dx + dy$$

on the additive group.

Remark 2.1. The identity (2.1) can be checked without developing a general theory of differentials on products. On the affine part of $E \times E$, use coordinates (x_1, y_1) and (x_2, y_2), and write $(x_3, y_3) = (x_1, y_1) + (x_2, y_2)$. The differential $\omega(x_3, y_3)$ can be expressed in the form

$$f(x_1, y_1, x_2, y_2)\, \omega(x_1, y_1) + g(x_1, y_1, x_2, y_2)\, \omega(x_2, y_2)$$

for suitable rational functions f and g. Fixing the second point and using translation invariance gives $f = 1$; fixing the first point gives $g = 1$. This proves (2.1) on a dense open set, and hence everywhere.

Recall that the pullback of a differential by a constant morphism is zero. Indeed, if c is a constant morphism, then

$$c^*(f\, dg) = (f \circ c)\, d(g \circ c) = 0.$$

In particular,

$$[0]^* \omega = 0.$$

Theorem 2.2. *Let E' and E be elliptic curves, let ω be an invariant differential on E, and let*

$$\phi, \psi : E' \longrightarrow E$$

be isogenies. Then

$$(\phi + \psi)^* \omega = \phi^* \omega + \psi^* \omega.$$

Proof. Consider the morphism

$$(\phi, \psi) : E' \longrightarrow E \times E, \qquad P \longmapsto (\phi(P), \psi(P)).$$

Since

$$\phi + \psi = \mu \circ (\phi, \psi),$$

pulling back (2.1) gives

$$\begin{aligned}(\phi+\psi)^*\omega &= (\phi,\psi)^*\mu^*\omega \\ &= (\phi,\psi)^*\left(\mathrm{pr}_1^*\,\omega + \mathrm{pr}_2^*\,\omega\right) \\ &= \phi^*\omega + \psi^*\omega.\end{aligned}$$

□

Corollary 2.3. *Let ω be an invariant differential on E. For every integer m,*

$$[m]^*\omega = m\omega.$$

Proof. The formula is clear for $m = 0$ and $m = 1$. The theorem gives

$$[m+1]^*\omega = [m]^*\omega + [1]^*\omega,$$

so induction proves the formula for $m \geq 0$. Taking $[m] + [-m] = [0]$ gives

$$[m]^*\omega + [-m]^*\omega = 0,$$

which gives the formula for negative m as well. □

In particular,

$$[-1]^*\omega = -\omega.$$

This is also immediate from the usual formula

$$[-1](x, y) = (x, -y - a_1x - a_3).$$

3. The differential of an isogeny

Let

$$\phi : E_1 \longrightarrow E_2$$

be an isogeny, and choose invariant differentials ω_1 and ω_2 on E_1 and E_2. Since pullback carries regular differentials to regular differentials and the space of regular differentials on E_1 is one dimensional, there is a unique scalar

$$a_\phi = a_\phi(\omega_1, \omega_2) \in \bar{K}$$

such that

$$\phi^*\omega_2 = a_\phi\omega_1. \tag{3.1}$$

For the zero isogeny, we have $a_{[0]} = 0$.

Remark 3.1. The scalar a_ϕ is the scalar by which ϕ acts on the cotangent spaces at the identity points. Thus it is the algebraic analogue of the derivative of a homomorphism of one-dimensional Lie groups at the identity.

If the curves, the isogeny, and the chosen differentials are all defined over K, then

$$a_\phi \in K.$$

This follows by applying every element of G_K to (3.1).

Proposition 3.2. *Let $\phi : E_1 \to E_2$ be a nonzero isogeny. Then the following are equivalent:*

(i) ϕ is separable;
(ii) $\phi^*\omega_2 \neq 0$;
(iii) $a_\phi \neq 0$.

Proof. We proved earlier for morphisms of smooth curves that a morphism is separable if and only if its pullback on rational differentials is nonzero. Since ω_2 spans the one-dimensional space of rational differentials over $\bar{K}(E_2)$, this is equivalent to

$$\phi^*\omega_2 \neq 0.$$

The equivalence with $a_\phi \neq 0$ follows from (3.1). □

Proposition 3.3. *Let*

$$E_1 \xrightarrow{\phi} E_2 \xrightarrow{\psi} E_3$$

be isogenies, and choose invariant differentials ω_i on E_i. Then

$$a_{\psi\circ\phi} = a_\psi a_\phi.$$

If $\phi, \psi : E_1 \to E_2$, then

$$a_{\phi+\psi} = a_\phi + a_\psi.$$

Proof. The first formula follows from functoriality of pullback:

$$\begin{aligned}(\psi\circ\phi)^*\omega_3 &= \phi^*(\psi^*\omega_3)\\ &= \phi^*(a_\psi\omega_2)\\ &= a_\psi a_\phi \omega_1.\end{aligned}$$

The second formula is the preceding theorem. □

Example 3.4 (The explicit 2-isogeny)**.** Assume that $\mathrm{char}(K) \neq 2$, and recall the isogeny from the preceding lecture

$$\phi : E \longrightarrow E',$$

where

$$E : y^2 = x^3 + ax^2 + bx,$$

$$E' : Y^2 = X^3 - 2aX^2 + (a^2 - 4b)X,$$

and

$$X = x + a + \frac{b}{x}, \qquad Y = \frac{y(b - x^2)}{x^2}.$$

The invariant differentials are

$$\omega_E = \frac{dx}{2y}, \qquad \omega_{E'} = \frac{dX}{2Y}.$$

Since

$$dX = \left(1 - \frac{b}{x^2}\right) dx = \frac{x^2 - b}{x^2}\, dx$$

and

$$2Y = -\frac{2y(x^2 - b)}{x^2},$$

we obtain

$$\phi^*\omega_{E'} = \frac{dX}{2Y} = -\frac{dx}{2y} = -\omega_E.$$

Thus

$$a_\phi = -1,$$

and the nonvanishing of this scalar again shows that ϕ is separable.

4. Multiplication maps and Frobenius

The invariant differential gives a quick separability criterion for multiplication maps.

Corollary 4.1. *Let E/K be an elliptic curve and let $m \in \mathbb{Z}$. If the image of m in K is nonzero, then*

$$[m] : E \longrightarrow E$$

is a nonzero separable endomorphism. In particular, if $m \neq 0$, then

$$[m] \text{ is separable} \quad \Longleftrightarrow \quad \mathrm{char}(K) \nmid m.$$

Thus every nonzero multiplication map is separable in characteristic zero.

Proof. We have

$$[m]^*\omega = m\omega.$$

If $m \neq 0$ in K, this differential is nonzero, so $[m]$ is nonzero and separable. If $\mathrm{char}(K) = p > 0$ and $p \mid m$, then

$$[m]^*\omega = 0,$$

so the nonzero map $[m]$ is inseparable. □

We next return to Frobenius. Let E be defined over the finite field $\mathbb{F}_q$, where $q = p^r$, and let

$$\pi : E \longrightarrow E, \qquad (x, y) \longmapsto (x^q, y^q)$$

be the q-power Frobenius endomorphism. Since the coefficients of a Weierstrass equation for $E/\mathbb{F}_q$ are fixed by the q-power map, the Frobenius twist is naturally the same curve E.

Proposition 4.2. *Let ω be an invariant differential on $E/\mathbb{F}_q$. Then*

$$\pi^*\omega = 0.$$

Consequently, for integers m, n, the endomorphism

$$[m] + [n] \circ \pi : E \longrightarrow E$$

is separable if and only if $p \nmid m$. In particular, $[1] - \pi$ is separable.

Proof. On a Weierstrass equation, the numerator of the invariant differential is dx. Therefore

$$\pi^*(dx) = d(x^q) = qx^{q-1}dx = 0$$

in characteristic p. Hence $\pi^*\omega = 0$.

Using additivity of pullback and the formula for multiplication maps, we obtain

$$\begin{aligned} ([m] + [n] \circ \pi)^*\omega &= [m]^*\omega + ([n] \circ \pi)^*\omega \\ &= m\omega + n\pi^*\omega \\ &= m\omega. \end{aligned}$$

The separability criterion now gives the result. □

Remark 4.3. A geometric point $P \in E(\bar{K})$ belongs to $E(\mathbb{F}_q)$ if and only if

$$\pi(P) = P.$$

Therefore

$$E(\mathbb{F}_q) = \ker([1] - \pi).$$

Since $[1] - \pi$ is separable, we obtain the useful identity

$$\#E(\mathbb{F}_q) = \deg([1] - \pi).$$

Later, the study of the degree function on $\mathrm{End}(E)$ will turn this identity into information about the number of points on E over finite fields.

5. The scalar action of the endomorphism ring

Theorem 5.1. *The map*

$$\partial : \mathrm{End}(E) \longrightarrow \bar{K}, \qquad \phi \longmapsto a_\phi,$$

is a ring homomorphism. Its kernel consists of the zero endomorphism together with the nonzero inseparable endomorphisms of E.

Proof. The addition theorem gives $a_{\phi+\psi} = a_\phi + a_\psi$, while functoriality of pullback gives $a_{\phi\circ\psi} = a_\phi a_\psi$. Also, $a_{[1]} = 1$ and $a_{[0]} = 0$. Thus ∂ is a ring homomorphism. The description of its kernel follows from the equivalent conditions for separability. □

Corollary 5.2. *If* $\mathrm{char}(K) = 0$*, then*

$$\mathrm{End}(E) \hookrightarrow \bar{K}$$

through the map $\phi \mapsto a_\phi$. Consequently, $\mathrm{End}(E)$ *is commutative.*

Proof. In characteristic zero every nonzero endomorphism is separable. Hence the kernel of ∂ is zero, so ∂ is injective. Its target is a commutative field, and therefore $\mathrm{End}(E)$ is commutative. □

Example 5.3. Assume that $\mathrm{char}(K) \neq 2$, let $i \in K$ satisfy $i^2 = -1$, and consider

$$E : y^2 = x^3 - x.$$

For the endomorphism

$$\iota(x, y) = (-x, iy),$$

we have

$$\iota^* \left(\frac{dx}{2y} \right) = \frac{d(-x)}{2iy} = i \frac{dx}{2y}.$$

Thus

$$a_\iota = i.$$

The scalar action on the invariant differential therefore reflects the relation

$$\iota^2 = [-1]$$

through the equality

$$i^2 = -1.$$